

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

A university database has a Person superclass with Student and Faculty subclasses. A person can be both a student (doing PhD) and faculty member (teaching). Additionally, not all persons in the database are students or faculty (some are visitors). Which ISA mapping option is MOST appropriate?

- A) Option 1 - Multiple relations with superclass and subclass tables
 - B) Option 2 - Separate relations for subclasses only
 - C) Option 3 - Single relation with type attribute
 - D) Option 4 - Single relation with Boolean attributes **Answer: A**
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Question 2

Consider the following relational schema:

Employee(eid, ename, salary, did)

Department(did, dname, budget)

You need to enforce: "Total salary of all employees in a department should not exceed the department's budget." Which constraint type is needed?

- A) Domain constraint using CHECK
 - B) Key constraint
 - C) Referential integrity using foreign key
 - D) Assertion (constraint across multiple tables) **Answer: D**
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Question 3

In mapping a composite attribute "Address" (street, city, postal_code) to the relational model, what is the correct approach?

- A) Create a separate Address table with foreign key reference
- B) Flatten into simple attributes: street, city, postal_code in the same relation
- C) Store as a single text field with concatenated values
- D) Use a multivalued attribute representation

Answer: B

Question 4

Given an M:N relationship "Enrolls" between Student and Course entities with a descriptive attribute "grade", how should this be mapped?

- A) Include grade as an attribute in Student table
 - B) Include grade as an attribute in Course table
 - C) Create an Enrolls table with Student PK, Course PK as composite key, and grade as attribute
 - D) Create two separate tables for the relationship **Answer: C**
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Question 5

A weak entity "Room" depends on strong entity "Building" through identifying relationship "Contains". Room has partial key "room_number" and Building has key "building_id". What is the primary key of the Room relation?

- A) room_number only
- B) building_id only
- C) (building_id, room_number) - composite key
- D) A new surrogate key should be created

Answer: C

Question 6

You have a ternary relationship "Supply" connecting Supplier, Part, and Project with attribute "quantity". How many foreign keys will the Supply relation contain?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C

Question 7

Consider ISA mapping Option 2 for Person → Student, Faculty. Which scenario creates a problem?

- A) A person who is only a student
- B) A person who is only a faculty member
- C) A person who is neither student nor faculty (visitor)
- D) A person who is both student and faculty

Answer: C

Question 8

In an Employee-Project-Department scenario, you have:

- Employee works_in Department (N:1)
- Department manages Project (1:N)
- Employee works_on Project (M:N)

If you try to determine which employees work on a project using only the works_in and manages relationships, what design trap occurs?

- A) Fan trap
- B) Chasm trap
- C) Circular trap
- D) No trap occurs

Answer: B

Question 9

A foreign key constraint "ON DELETE CASCADE" means:

- A) Deletion is not allowed if referenced tuples exist
- B) When a referenced tuple is deleted, all referencing tuples are also deleted
- C) When a tuple is deleted, its foreign key is set to NULL
- D) Deletion creates a backup before removing data

Answer: B

Question 10

In ISA mapping Option 3 (single table with type attribute), you have Person with Student and Faculty subclasses. Student has "gpa" and Faculty has "salary". A visitor (neither student nor faculty) would have:

- A) Type = 'Visitor', gpa = NULL, salary = NULL
- B) Cannot be represented in this option
- C) Type = NULL, gpa = NULL, salary = NULL
- D) Requires a separate table

Answer: A

Question 11

You are mapping an aggregation: Players play Matches, and this play relationship wins Awards. What is the primary key of the Award relation?

- A) awardName only
- B) (playerId, matchId) only
- C) (awardName, playerId, matchId) - composite of all three
- D) (playerId, matchId, awardDate)

Answer: C

Question 12

Which of the following violations would a domain constraint catch?

- A) An employee's salary exceeding their manager's salary
- B) Entering age as -5 for a student
- C) A foreign key referencing a non-existent primary key
- D) Duplicate primary key values

Answer: B

Question 13

Consider a 1:1 relationship "Manages" between Employee and Department where each department has exactly one manager and an employee manages at most one department. What is the MOST efficient mapping strategy?

- A) Create a separate Manages relation
- B) Add department_id as foreign key in Employee table
- C) Add employee_id as foreign key in Department table with NOT NULL
- D) Merge Employee and Department into one table **Answer: C**

Question 14

In a university database, you have ISA hierarchy: Person → Employee → Faculty/Staff (disjoint, total). Best mapping approach?

- A) Option 1 - Four tables: Person, Employee, Faculty, Staff
- B) Option 2 - Two tables: Faculty, Staff (with Person and Employee attributes)
- C) Option 3 - One table with type attribute
- D) Multiple hierarchy mapping

Answer: B

Question 15

A candidate key that is NOT chosen as the primary key is called:

- A) Foreign key
- B) Alternate key
- C) Composite key
- D) Partial key

Answer: B

Question 16

You're converting a multivalued attribute "phone_numbers" for Employee entity. What is the correct approach?

- ☐
- ☐
- ☐
- ☐

- A) Store as comma-separated values in one column
- B) Create columns phone1, phone2, phone3
- C) Create a separate Phone table (emp_id, phone_number) with composite PK
- D) Store in JSON format

Answer: C

Question 17

Which ISA mapping option would have the MOST NULL values when subclasses have many unique attributes?

- A) Option 1
- B) Option 2
- C) Option 3
- D) Option 4

Answer: D

Question 18

In a Book-Author relationship where a book can have multiple authors and an author can write multiple books, with an attribute "author_order" (1st author, 2nd author, etc.), the primary key of the relationship relation is:

- A) (book_id, author_id)
- B) author_order
- C) (book_id, author_order)
- D) (book_id, author_id, author_order)

- ☐
- ☐
- ☐
- ☐

Answer: A

Question 19

A fan trap occurs when:

- A) Too many relationships originate from one entity causing ambiguous paths
- B) A relationship is missing between entities
- C) An entity has too many attributes
- D) Foreign key references are circular

Answer: A

Question 20

Referential integrity "ON DELETE SET NULL" is NOT appropriate when:

- A) The foreign key is part of the primary key
- B) The relationship is optional
- C) Cascade deletion is expensive
- D) The referenced table is large

Answer: A

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) Question 21

When mapping a 1:N relationship between Department (1) and Employee (N), which approaches are valid? **(Select all that apply)**

- A) Add department_id as FK in Employee table
- B) Add employee_id as FK in Department table

- ☐
- ☐
- ☐
- ☐
- C) Create a separate DeptEmp relation
- D) Include relationship attributes in Employee table
- E) Merge both entities into one table

Answers: A, C, D

Question 22

A legal database instance must satisfy: **(Select all that apply)**

- A) All domain constraints
- B) All key constraints (no duplicate PKs)
- C) All referential integrity constraints
- D) All business rules implemented as assertions
- ☐ E) Minimum storage requirements

Answers: A, B, C, D

Question 23

In ISA mapping Option 1, which statements are TRUE? **(Select all that apply)**

- A) Allows querying superclass without JOIN operations
 - B) Efficiently handles overlapping subclasses
 - C) Requires NULL values for subclass-specific attributes
 - D) Each subclass relation's PK is also a FK to superclass
 - E) Cannot represent entities that belong to no subclass **Answers: B, D**
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Question 24

Which factors should influence the choice of ISA mapping option? **(Select all that apply)**

- A) Whether specialization is overlapping or disjoint
- B) Whether specialization is total or partial

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- ☐
- ☐
- ☐
- ☐
- C) Number of attributes in subclasses
 - D) Query patterns (frequent superclass vs subclass queries)
 - E) Number of tuples expected

Answers: A, B, C, D

Question 25

When mapping aggregation relationships, which are correct? **(Select all that apply)**

- A) First map the inner aggregation relationship
- B) The aggregation is treated as an entity for outer relationship
- C) Primary key of aggregation table combines PKs of aggregated entities

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D) Outer relationship's FK references the aggregation table's PK

E) Aggregation eliminates the need for ternary relationships **Answers: A,**

B, C, D

Question 26

For a weak entity to be correctly mapped: **(Select all that apply)**

- A) Include owner entity's PK as FK
- B) Combine owner's PK with partial key for weak entity's PK
- C) Weak entity can exist independently in the database
- D) Owner's PK must be part of weak entity's primary key
- E) Create separate table for the identifying relationship **Answers: A, B, D**

Question 27

CHECK constraints can be used to enforce: **(Select all that apply)**

- A) Salary > 0
- B) Age BETWEEN 18 AND 65
- C) Start_date < End_date
- D) Employee salary < Manager salary (different tuples)
- E) Email format validation

Answers: A, B, C, E

Question 28

In ISA mapping Option 4 with Boolean flags, problems include: **(Select all that apply)**

- A) Many NULL values for subclass-specific attributes
- B) No way to enforce disjoint constraint at schema level

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- C) Difficult to add new subclasses
- D) Cannot represent overlapping subclasses
- E) Poor performance for subclass-specific queries

Answers: A, B, C, E

Question 29

Which relationships require a separate relation table? **(Select all that apply)**

- A) M:N binary relationship
- B) 1:N binary relationship with relationship attributes
- C) Ternary relationship
- D) 1:1 relationship with total participation on both sides
- E) Aggregation relationship

Answers: A, B, C

Question 30

Foreign key ON DELETE options and their effects: **(Select all that apply)**

- A) CASCADE - deletes all referencing tuples
- B) SET NULL - sets FK to NULL in referencing tuples
- C) RESTRICT - prevents deletion if references exist
- D) SET DEFAULT - sets FK to default value
- E) DO NOTHING - allows orphaned records

Answers: A, B, C, D

SECTION C: SCENARIO-BASED QUESTIONS Question 31

Scenario: Hospital database has Doctor and Patient entities. A doctor treats many patients; a patient is treated by many doctors. Treatment has "start_date" and "diagnosis". Some

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doctors are specialists with "specialization" attribute; some are general practitioners. All doctors must be one or the other (total, disjoint).

What is the complete correct mapping?

- A) Doctor(doc_id, name), Specialist(doc_id, specialization), GP(doc_id), Patient(pat_id, name), Treats(doc_id, pat_id, start_date, diagnosis)
- B) Doctor(doc_id, name, specialization, type), Patient(pat_id, name), Treats(doc_id, pat_id, start_date, diagnosis)
- C) Specialist(doc_id, name, specialization), GP(doc_id, name), Patient(pat_id, name), Treats(doc_id, pat_id, start_date, diagnosis)
- D) Doctor(doc_id, name, specialization, is_specialist), Patient(pat_id, name), Treats(doc_id, pat_id, start_date, diagnosis)

Answer: C

Question 32

Scenario: Library system: Book has ISBN, title. Copy has copy_id, ISBN. Member has member_id. A copy is borrowed by at most one member; a member can borrow many copies. Borrowing has "due_date".

Which mapping is most efficient?

- A) Book(ISBN, title), Copy(copy_id, ISBN*), Member(member_id), Borrows(copy_id, member_id, due_date)
- B) Book(ISBN, title), Copy(copy_id, ISBN*, member_id*, due_date), Member(member_id)
- C) Book(ISBN, title), Copy(copy_id, ISBN*), Member(member_id, copy_id*, due_date)
- D) Book(ISBN, title, copy_id), Copy(copy_id, member_id, due_date), Member(member_id)

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Answer: B

Question 33

Scenario: University has Course, Instructor, Semester. An instructor teaches a course in a specific semester (same course can be taught by different instructors in different semesters). Each offering has "enrollment_count".

What is the correct mapping?

A) Course(course_id, name), Instructor(inst_id, name), Semester(sem_id, term),
Teaches(course_id, inst_id, sem_id, enrollment_count) with PK (course_id, sem_id)

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- B) Course(course_id, name), Instructor(inst_id, name), Semester(sem_id, term), Teaches(course_id, inst_id, sem_id, enrollment_count) with PK (course_id, inst_id, sem_id)
- C) Course(course_id, name, inst_id*, sem_id*), Instructor(inst_id, name), Semester(sem_id, term)
 - D) CourseOffering(course_id, inst_id, sem_id, enrollment_count, name) with PK (course_id, sem_id)

Answer: B

Question 34

Scenario: Employee entity has emplID, name, salary, supervisor_ID (an employee's supervisor is also an employee - recursive relationship). How should this be mapped?

- A) Employee(emplID, name, salary), Supervises(emplID, supervisor_ID)
- B) Employee(emplID, name, salary, supervisor_ID*)
- C) Employee(emplID, name, salary), Supervisor(supervisor_ID, emplID*)
- D) Cannot be mapped in relational model

Answer: B

Question 35

Scenario: Person → Employee → Manager/Engineer (overlapping, partial). Person has NIC, name. Employee has emplID, salary. Manager has dept_name. Engineer has specialization. Some employees are both manager and engineer.

Which option prevents excessive NULL values while supporting all constraints?

- A) Option 1 - Four tables: Person, Employee, Manager, Engineer
- B) Option 2 - Two tables: Manager, Engineer
- C) Option 3 - One table with type attribute
- D) Option 4 - One table with Boolean flags

Answer: A

Question 36

Which of the following is true related to ISA relationships? (Select one or more)

- A) The hierarchy in which each entity participates in only one subclass relationship is classified as disjoint
- B) A subtype of an ISA hierarchy can have its own key
- C) A shape object with circle, rectangle, and triangle as subtypes is an example of total participation
- D) If subtypes are overlapping, the participation constraint must be partial

Answer: A, B, D

Question 37

Which of the following is/are TRUE related to ISA relationships? (Select one or more)

- A) The hierarchy in which each entity participates in only one subclass relationship is classified as disjoint
- B) A subtype of an ISA hierarchy can have its own key
- C) A shape object with circle, rectangle, and triangle as subtypes is an example of total participation
- D) If subtypes are overlapping, the participation constraint must be partial

Answer: A, B, D

Question 38

Construction company – EER diagram (binary/ternary) Correct statements:

- A) Binary relationship between site and material with descriptive attribute
- B) Only binary relationships in the diagram
- C) Company is a strong entity in the EER
- D) Ternary relationship exists
- E) Raw material types could be represented by subclasses

Answer: A, C, D, E

Question 39

ISA relationships Which are correct?

- A) Subtypes at a higher level inherit attributes only from immediate subtype
- B) Defining supertypes and forming supertype/subtype relationships is called specialization
- C) ISA with private university, state university, semi-government university is partial and disjoint

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- D) A subtype can participate in a relationship unique to that subtype

Answer: B, D

Question 40

University rooms scenario: A university has lecture halls and laboratories. Lecture halls have a capacity and resources such as whiteboard, podium, and projector. Labs have capacity and resources. Rooms are in buildings (Block A, B, C). Each room has a number unique to each building. Batches are identified by intake year and name (e.g., 2021 Regular Intake). Groups exist within batches, each group has a unique name within the batch. Rooms are allocated for classes with a class name, start time, and end time. Which of the following are true?

- A) A group cannot exist without a batch
- B) There are no descriptive attributes for relationships in the diagram
- C) There are two identifying relationships in the diagram
- D) Allocated can be considered as a ternary relationship
- E) The entity batch has a composite key

Answer: A, C, E

Question 41

Construction company – EER diagram Which of the following are correct related to the EER diagram?

- A) Raw material types could be represented by an attribute named type
- B) Site is involved with a binary relationship with raw material
- C) Company is a strong entity in the EER
- D) Purchase relationship contains three descriptive attributes
- E) Site is involved in a ternary relationship

Answer: C, E

Question 42

Consider a relation $R(A,B,C,D,E,F,G)$ with the following set of functional dependencies: $F = \{ AB \rightarrow CDEFG, D \rightarrow B, D \rightarrow E, E \rightarrow FG \}$. The corresponding BCNF relations are:

- A) $R_1(D,E)$, $R_2(A,B,C,D)$
- B) $R_1(E,F,G)$, $R_2(A,B,C,D,E)$
- C) $R_1(E,F,G)$, $R_2(D,E)$, $R_3(A,B,C,D)$
- D) $R_1(A,B,C,D,E,F,G)$
- E) $R_1(E,F,G)$, $R_2(D,E,B)$, $R_3(A,C,D)$

Answer: C

Question 43

Company has multiple construction sites. Each site has a unique site number, address, and construction type such as 'house', 'apartment', 'shopping complex'. For each site, the company estimates the number and amounts required from each raw material. Raw materials have three different types: wall-construction materials, wiring materials, roofing materials, and timber materials. Each material is identified by a unique ID and has a unit of measurement such as liters and kilograms. There are many suppliers providing raw materials. Suppliers are identified by a unique supplier ID and each supplier has a name, address, and phone number. During construction, the company may purchase different amounts of raw materials for a site from different suppliers at different prices. Which of the following are correct related to the EER diagram drawn for the above description? (Select one or more)

- A) Raw material types could be represented by an attribute named type
- B) Site is involved with a binary relationship with raw material
- C) Company is a strong entity in the EER
- D) Purchase relationship contains three descriptive attributes
- E) Site is involved in a ternary relationship

Answer: C, E

Question 44

Select the correct answer(s) about the 'Section' relation in the final relational model. (Select one or more)

- A) Section relation has a composite primary key
- B) Bookid is a foreign key in the Section relation
- C) The degree of the Section relation is 2
- D) secno is the primary key of the Section relation
- E) The degree of the Section relation is 3

Answer: A, B

Question 45

Consider the following requirements for a construction company:

- The company has multiple construction sites. Each site has a unique site number, address, and construction type (e.g., 'house', 'apartment', 'shopping complex').
- For each site, the company estimates the amounts required from each raw material (e.g., cement packs, sand, etc.).
- Raw materials have three different types: wall-construction materials, wiring materials, roofing materials, and timber materials. Each material is identified by a unique ID and has a unit of measurement (e.g., liters, kilograms).
- There are many suppliers providing raw materials. Suppliers are identified by a unique supplier ID and each has a name, company address, and phone number.

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• During construction, different amounts of raw materials for a site may come from different suppliers at different prices.

Which of the following are correct related to the EER diagram drawn for the above description? (Select one or more)

- A) There is a ternary relationship in the diagram
- B) There are only binary relationships in the diagram
- C) There is a binary relationship between site and material with a descriptive attribute
- D) Raw material types could be represented by subclasses
- E) Company is a strong entity in the EER

Answer: A, C, D, E

Question 46

Consider the following relational schemes: Car(Number, Owner, ChassisNo, Model, Year, Price) Registration(Number, Owner, ChassisNo) With the following functional dependencies:

1. Number, Owner $\rightarrow$ ChassisNo
2. ChassisNo $\rightarrow$ Number, Owner, Model, Year
3. Model, Number, Year $\rightarrow$ Price

Assume {Number, Owner} is the key for both schemes. Question: What is the current normal form of Registration?

- A) 3NF
- B) BCNF
- C) Unnormalized
- D) 2NF
- E) 1NF

Answer: B

Question 47

Let R(A,B,C) be a relation with the functional dependencies $A \rightarrow B$ and $B \rightarrow C$. Which of the following is a candidate key?

- A) A
- B) B
- C) C
- D) A and B
- E) A and C

Answer: A

Question 48

The lossless decomposition of relation R is ensured if and only if:

- A) Every functional dependency is preserved
- B) Common attribute(s) form a superkey in at least one relation
- C) Common attribute(s) are candidate keys in both relations
- D) Decomposition is done in BCNF
- E) All relations are 3NF

Answer: B

Question 49

Which of the following statements about 3NF and BCNF is true?

- A) Every BCNF relation is in 3NF
- B) Every 3NF relation is in BCNF
- C) BCNF allows transitive dependencies
- D) 3NF eliminates all anomalies
- E) BCNF allows redundancy

Answer: A

Question 50

A functional dependency $X \rightarrow Y$ holds in a relation R if and only if:

- A) X is a superkey
- B) Y is a key
- C) Each value of X is associated with precisely one value of Y
- D) Y determines X
- E) X and Y are disjoint sets

Answer: C

Question 51

Which of the following normal forms is based on the concept of functional dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF
- E) All of the above

Answer: E

Question 52

Candidate keys – CustomerSales CustomerSales(CustNo, SalesDate, SalesAmount, SalesRepNo, Location) FDs:

- CustNo, SalesDate -> SalesAmount, SalesRepNo, Location
- SalesRepNo, SalesDate, SalesTime -> CustNo
- Location, SalesDate, SalesTime -> SalesRepNo, CustNo

Candidate keys:

- A) (Location, SalesDate, SalesTime)
- B) (CustNo, SalesDate)
- C) CustNo, SalesRepNo
- D) SalesRepNo, SalesDate, SalesTime
- E) CustNo

Answer: B, D, E

Question 53

Select the correct answer(s) about the 'Section' relation in the final relational model. (Select one or more)

- A) Section relation has a composite primary key
- B) Bookid is a foreign key in the Section relation
- C) The degree of the Section relation is 2
- D) secno is the primary key of the Section relation
- E) The degree of the Section relation is 3

Answer: A, B

Question 54

- a) It is in 1NF and every non-prime attribute is fully functionally dependent on the primary key
- b) There are no transitive dependencies
- c) Every determinant is a candidate key
- d) There is no partial dependency
- e) a and d only

Answer: B

Question 55

Which of the following causes anomalies in databases?

- A) Redundancy
- B) Lack of normalization
- C) Improper constraints
- D) All of the above
- E) None of the above

Answer: D

Question 56

The primary purpose of normalization is to:

- A) Reduce redundancy and avoid anomalies
- B) Increase query speed
- C) Ensure data security
- D) Improve indexing
- E) Facilitate distributed databases

Answer: A

Question 57

Which of the following decompositions is not dependency preserving?

- A) $R(A,B,C) \rightarrow R_1(A,B)$ and $R_2(B,C)$ for $A \rightarrow B$ and $B \rightarrow C$
- B) $R(A,B,C) \rightarrow R_1(A,B)$ and $R_2(A,C)$ for $A \rightarrow B$ and $A \rightarrow C$
- C) $R(A,B,C) \rightarrow R_1(A,B)$ and $R_2(B,C)$ for $A \rightarrow C$
- D) $R(A,B,C) \rightarrow R_1(B,C)$ and $R_2(C,A)$ for $A \rightarrow B$
- E) None of these

Answer: C

Question 58

Consider a relation $R(A, B, C, D, E, F, G)$ with the following set of functional dependencies: $F = \{ AB \rightarrow CDEFG, D \rightarrow B, D \rightarrow E, E \rightarrow FG \}$. The corresponding BCNF relations are:

- A) $R_1(D, E), R_2(A, B, C, D)$
- B) $R_1(E, F, G), R_2(A, B, C, D, E)$
- C) $R_1(E, F, G), R_2(D, E), R_3(A, B, C, D)$
- D) $R_1(A, B, C, D, E, F, G)$
- E) $R_1(E, F, G), R_2(D, E, B), R_3(A, C, D)$

Answer: C

Question 59

Consider a relation $R(A, B, C, D, E, F, G)$ with functional dependencies: $F = \{ AB \rightarrow C, A \rightarrow D, G \rightarrow A, B \rightarrow EFG \}$. The corresponding BCNF relations are:

- A) $R_1(A, D), R_2(B, E, F, G), R_3(A, B, C)$
- B) $R_1(A, D), R_2(B, E, F, G), R_3(A, B, C, G)$
- C) $R_1(A, D), R_2(B, E, F, G), R_3(A, G), R_4(B, C, G)$
- D) $R_1(A, D), R_2(B, E, F, G), R_3(A, B, C), R_4(A, G)$
- E) $R_1(A, B, C, D, E, F, G)$

Answer: D

Question 60

If a view named V1 is created that references every column in table EMPLOYEE except the column SALARY, which of the following is not a correct statement? (Select one or more)

- A) All data, except SALARY data, that is stored in the EMPLOYEE table is copied to a separate physical location when V1 is created.
- B) View V1 can be used in the same context as the EMPLOYEE table for all data retrieval operations that do not require SALARY information.
- C) View V1 cannot be used as a data source for other views.
- D) No data is copied to a separate physical location when V1 is created.
- E) View V1 can be used as a data source for other views.

Answer: A, C

Question 61

Consider relation $R(F1, F2, F3, F4, F5)$ with functional dependencies: $F = \{ F2 F3 \rightarrow F1 F4 F5, F4 \rightarrow F2 \}$. What is the highest normal form of R?

- A) 3rd Normal Form (3NF)
- B) 2nd Normal Form (2NF)
- C) 1st Normal Form (1NF)
- D) Unnormalized Form
- E) BCNF

Answer: E

Question 62

Consider the following relation: $R(A, C, E, G, I, K, L, N, P, Q)$ With the following set of functional dependencies: $\{ACE, AGI, CK, K \rightarrow LN, GPQ\}$ Question: Identify the candidate key(s) in relation R. (Select one or more)

- A) A
- B) G
- C) C
- D) AC
- E) K

Answer: A, B

Question 63

Consider the following relational schemes: Car(Number, Owner, ChassisNo, Model, Year, Price) Registration(Number, Owner, ChassisNo) With the following functional dependencies:

1. Number, Owner $\rightarrow$ ChassisNo
2. ChassisNo $\rightarrow$ Number, Owner, Model, Year
3. Model, Number, Year $\rightarrow$ Price

Assume {Number, Owner} is the key for both schemes. What is the current normal form of Registration?

- A) 3NF
- B) BCNF
- C) Unnormalized
- D) 2NF
- E) 1NF

Answer: B

Question 64

Consider the table Academic_Staff(SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) with functional dependencies:

- $SID \rightarrow StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone$
- $FacultyID \rightarrow FacultyLocation, FacultyPhone$
- $FacultyLocation \rightarrow FacultyID, FacultyPhone$
- $FacultyPhone \rightarrow FacultyID, FacultyLocation$

What is the current normal form of Academic_Staff?

- A) Unnormalized Form
- B) 1NF
- C) 2NF
- D) 3NF
- E) BCNF

Answer: C

Question 65

Which of the following statements about rooms in a university description are true? (Select one or more)

- A) The type of the rooms could be represented using subclasses.
- B) The type of the rooms could be represented as an attribute.
- C) There are only strong entities in this description.
- D) If building is an entity, room will be a weak entity.
- E) Resources could be tracked using a multi-valued attribute.

Answer: A, B, D, E

Question 66

University rooms description: lecture halls and laboratories—both have capacity and number of resources; located in buildings (e.g., Block A, Block B); each room has a number unique within each building; batches identified by intake year and intake name; groups unique within batch; groups have number of students; timetable records class name, start and end time. Which of the following are true related to this description? (Select one or more)

- A) The type of the rooms could be represented using subclasses.
- B) The type of the rooms could be represented as an attribute.
- C) There are only strong entities in this description.
- D) If building is an entity, room will be a weak entity.
- E) Resources could be tracked using a multi-valued attribute.

Answer: A, B, D, E

Question 67

Which option below is the most appropriate mapping for an ISA hierarchy?

- A) Option 1 and Option 4
- B) Option 4
- C) Option 2
- D) Option 3
- E) None of the above

Answer: B

Question 68

Which of the following is true related to ISA relationships? (Select one or more)

- A) A shape object with circle, rectangle, and triangle as subtypes is an example of total participation.
- B) A subtype of an ISA hierarchy can have its own key.
- C) The hierarchy in which each entity participates in only one subclass relationship is classified as disjoint.
- D) If subtypes are overlapping, the participation constraint must be partial.

Answer: B, C, D

Question 69

Consider relation $R(M, N, O, P, Q)$ with functional dependencies: $F = \{ N \rightarrow M, M \rightarrow O, NO \rightarrow P, MO \rightarrow NQ \}$. What is the current normal form of R ?

- A) BCNF
- B) 1NF
- C) 2NF
- D) 3NF
- E) Unnormalized Form

Answer: D

Question 70

Consider a relation R with eight attributes: R, U, S, Q, T, P (atomic values only). Let the following set of functional dependencies hold for R : $F = \{ R \rightarrow S, Q \rightarrow R, U, T \rightarrow P \}$ Question: What is the normal form that the above relation is in?

- A) 2NF
- B) BCNF
- C) 1NF
- D) 3NF
- E) Unnormalized Form

Answer: A

Question 71

Consider the following relational schemes: $Car(Number, Owner, ChassisNo, Model, Year, Price)$
 $Registration(Number, Owner, ChassisNo)$ With the following functional dependencies:

1. $Number, Owner \rightarrow ChassisNo$
2. $ChassisNo \rightarrow Number, Owner, Model, Year$
3. $Model, Number, Year \rightarrow Price$

□

Assume {Number, Owner} is the key for both schemes. Question: What is the current normal form of Registration?

- A) BCNF
- B) 1NF
- C) Unnormalized
- D) 2NF
- E) 3NF

Answer: B

Question 72

Construction company scenario: multiple construction sites with unique site numbers, addresses, and types (house, apartment, shopping complex). Each site estimates amounts of raw materials (cement, sand, etc.) required. Raw materials have types (wall-construction, wiring, roofing, timber) and unique IDs with units. Multiple suppliers provide materials with unique supplier IDs, name, address, and phone. Purchases may vary by site, supplier, amount, and price. Which of the following are correct regarding the EER diagram? (Select one or more)

- A) There are only binary relationships in the diagram.
- B) There is a ternary relationship in the diagram.
- C) Company is a strong entity in the EER.
- D) There is a binary relationship between site and material with a descriptive attribute.
- E) Raw material types could be represented by subclasses.

Answer: B, C, D, E

Question 73

Consider relation $R(A, C, E, G, I, K, L, N, P, Q)$ with functional dependencies: $F = \{ ACE \rightarrow A, A \rightarrow G, CK \rightarrow K, K \rightarrow LN, G \rightarrow PQ \}$ Identify candidate keys in the relation R . (Select one or more)

- A) AC
- B) G
- C) C
- D) A
- E) K

Answer: A, B

Question 74

Consider relational schemes:

- Car(Number, Owner, ChassisNo, Model, Year, Price)
- Registration(Number, Owner, ChassisNo)

Functional dependencies:

1. Number, Owner $\rightarrow$ ChassisNo
2. ChassisNo $\rightarrow$ Number, Owner, Model, Year
3. Model, Number, Year $\rightarrow$ Price

Assume (Number, Owner) is the key for both schemes. What is the current normal form of Car?

- A) 1NF
- B) 2NF
- C) 3NF
- D) Unnormalized Form
- E) BCNF

Answer: C

Question 75

Academic_Staff(SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) with functional dependencies: $SID \rightarrow StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone$ $FacultyID \rightarrow FacultyLocation, FacultyPhone$ $FacultyLocation \rightarrow FacultyID, FacultyPhone$ $FacultyPhone \rightarrow FacultyID, FacultyLocation$ What is the primary key for the relation?

- A) SID
- B) SID, FacultyID
- C) FacultyPhone
- D) FacultyLocation
- E) StaffName

Answer: A

Question 76

Relation R has eight attributes R(U, S, Q, T, P). Consider R contains only atomic values. Functional dependencies: $F = \{ R \rightarrow S, Q \rightarrow RU, T \rightarrow P \}$ Identify candidate keys in the relation R. (Select one or more)

- A) QT
- B) T
- C) RQ
- D) Q
- E) R

Answer: A

Question 77

Relation R(U, S, Q, T, P) with atomic values. Functional dependencies: $F = \{ R \rightarrow S, Q \rightarrow RU, T \rightarrow P \}$ What is the normal form of this relation?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF
- E) Unnormalized Form

Answer: B

Question 78

Relation R(R, M, N, O, P, Q) with FDs: $F = \{ N \rightarrow M, M \rightarrow O, NO \rightarrow P, MO \rightarrow NQ \}$ Identify the candidate keys:

- A) N
- B) M
- C) MO
- D) NO
- E) O

Answer: D

Question 79

Which of the following is true related to ISA relationships? (Select one or more)

- A) The hierarchy in which each entity participates in only one subclass relationship is classified as disjoint.
- B) A subtype of an ISA hierarchy can have its own key.
- C) If subtypes are overlapping, the participation constraint must be partial.
- D) A shape object with circle, rectangle, and triangle as subtypes is an example of total participation.

Answer: A, B, C

Question 80

Relation R has eight attributes: R, U, S, Q, T, P. Consider relation R contains only atomic values. Let $F = \{ R \rightarrow S, Q \rightarrow RU, T \rightarrow P \}$ be a set of functional dependencies that hold for R. What is the normal form that the above relation is in?

- A) 3NF
- B) Unnormalized Form

- C) 1NF
- D) 2NF
- E) BCNF

Answer: D

Question 81

Academic Staff(SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) with functional dependencies: SID → StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone FacultyID → FacultyLocation, FacultyPhone FacultyLocation → FacultyID, FacultyPhone FacultyPhone → FacultyID, FacultyLocation What is the current normal form of Academic Staff?

- A) 1NF
- B) 2NF
- C) BCNF
- D) Unnormalized form
- E) 2NF

Answer: B

Question 82

Normal form Relation R(R, U, S, Q, T, P), F = {R → S, Q → RU, T → P} What is the normal form of R?

- A) 3NF
- B) Unnormalized Form
- C) 1NF
- D) 2NF
- E) BCNF

Answer: D

Question 83

Academic Staff – Normal form AcademicStaff(SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) FDs:

- SID → StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone
- FacultyID → FacultyLocation, FacultyPhone
- FacultyLocation → FacultyID, FacultyPhone
- FacultyPhone → FacultyID, FacultyLocation

Normal form of AcademicStaff:

- A) 1NF
- B) 2NF

□

- C) BCNF
- D) Unnormalized form
- E) 2NF

Answer: B

Question 84

Highest normal form – Relation R(F1–F5) Relation R(F1, F2, F3, F4, F5), FDs: F = {F2F3 -> F1F4F5, F4 -> F2} Highest normal form:

- A) 3rd Normal Form
- B) 2nd Normal Form
- C) 1st Normal Form
- D) Unnormalized Form
- E) BCNF

Answer: E

Question 85

Construction company – EER diagram permissions Scenario: Bank DB, users: Asanka and Ferosh (data entry operators) Which permissions are needed:

- A) ALTER ROLE dataEntry ADD MEMBER asanka.n
- B) ALTER ROLE db_owner ADD MEMBER ferosh.a
- C) GRANT INSERT on bankDB to dataEntry
- D) ALTER ROLE dataEntry ADD MEMBER ferosh.a
- E) ALTER ROLE db_owner ADD MEMBER asanka.n

Answer: A, C, D

Question 86

Candidate keys – Relation R(F1–F5) Relation R(F1, F2, F3, F4, F5), FDs: F = {F2F3 -> F1F4F5, F4 -> F2} Candidate keys:

- A) (F2F3, F4)
- B) {F4}
- C) {F2F3, F4F3}
- D) {F4F3}
- E) (F2F3)

Answer: B

Question 87

Relation R(A, B, C, D), FDs: F = {C → BD, D → A, A → C} Highest normal form:

- A) Unnormalized Form
- B) 1st Normal Form
- C) 3rd Normal Form
- D) BCNF
- E) 2nd Normal Form

Answer: C

Question 88

Consider the following relation: CustomerSales(CustNo, SalesDate, SalesAmount, SalesRepNo, Location) with the following set of functional dependencies:

1. CustNo, SalesDate → SalesAmount, SalesRepNo, Location
2. SalesRepNo, SalesDate, SalesTime → CustNo
3. Location, SalesDate, SalesTime → SalesRepNo, CustNo

Task: Identify the candidate keys of the relation CustomerSales. (Select one or more)

- A) (CustNo, SalesRepNo)
- B) (CustNo, SalesDate)
- C) (CustNo)
- D) (SalesRepNo, SalesDate, SalesTime)
- E) (Location, SalesDate, SalesTime)

Answer: B, D, E

Question 89

BCNF decomposition – Relation R(A–G) Relation R(A, B, C, D, E, F, G), FDs: F = {AB → CDEFG, D → B, D → E, E → FG} BCNF relations:

- A) R1(D, E), R2(A, B, C, D)
- B) R1(E, F, G), R2(A, B, C, D, E)
- C) R1(E, F, G), R2(D, E), R3(A, B, C, D)
- D) R1(A, B, C, D, E, F, G)
- E) R1(E, F, G), R2(D, E, B), R3(A, C, D)

Answer: C

Question 90

Normal form – Relation R(A–H) Relation R(A, B, C, D, E, F, G, H), FDs: F = {ABC → DE, D → B, E → FG, G → F} Highest normal form:

- A) BCNF

□

- B) 2nd Normal Form
- C) 1st Normal Form
- D) 3rd Normal Form
- E) Unnormalized Form

Answer: D

Question 91

BCNF decomposition – Relation R(A–G) Relation R(A, B, C, D, E, F, G), FDs: F = {AB → C, A → D, G → A, B → EFG} BCNF relations:

- A) R1(A, D), R2(B, E, F, G), R3(A, B, C)
- B) R1(A, D), R2(B, E, F, G), R3(A, B, C, G)
- C) R1(A, D), R2(B, E, F, G), R3(A, G), R4(B, C, G)
- D) R1(A, D), R2(B, E, F, G), R3(A, B, C), R4(A, G)
- E) R1(A, B, C, D, E, F, G)

Answer: D

Question 92

Candidate keys – Relation R(F1–F4) Relation R(F1, F2, F3, F4), FDs: F = {F3F4 → F2, F2 → F1, F1 → F3} Candidate keys:

- A) F1F4, F2F4
- B) F1F4
- C) F2F4
- D) F1F4, F2F4, F3F4
- E) F3F4

Answer: A

Question 93

Consider the following relation: CustomerSales(CustNo, SalesDate, SalesAmount, SalesRepNo, Location) With the following set of functional dependencies:

1. CustNo, SalesDate → SalesAmount, SalesRepNo, Location
2. SalesRepNo, SalesDate, SalesTime → CustNo
3. Location, SalesDate, SalesTime → SalesRepNo, CustNo

Identify candidate keys in the relation R. (Select one or more)

- A) (CustNo, SalesRepNo)
- B) (CustNo, SalesDate)
- C) (CustNo)
- D) (SalesRepNo, SalesDate, SalesTime)
- E) (Location, SalesDate, SalesTime)

Answer: B, D, E
